DSP

Forever Notes

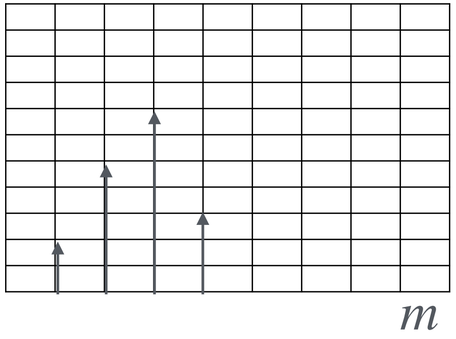
HW

Chapter 3

DFT

$$Analysis\ Frequencies = \frac{m \cdot f_s}{N}$$

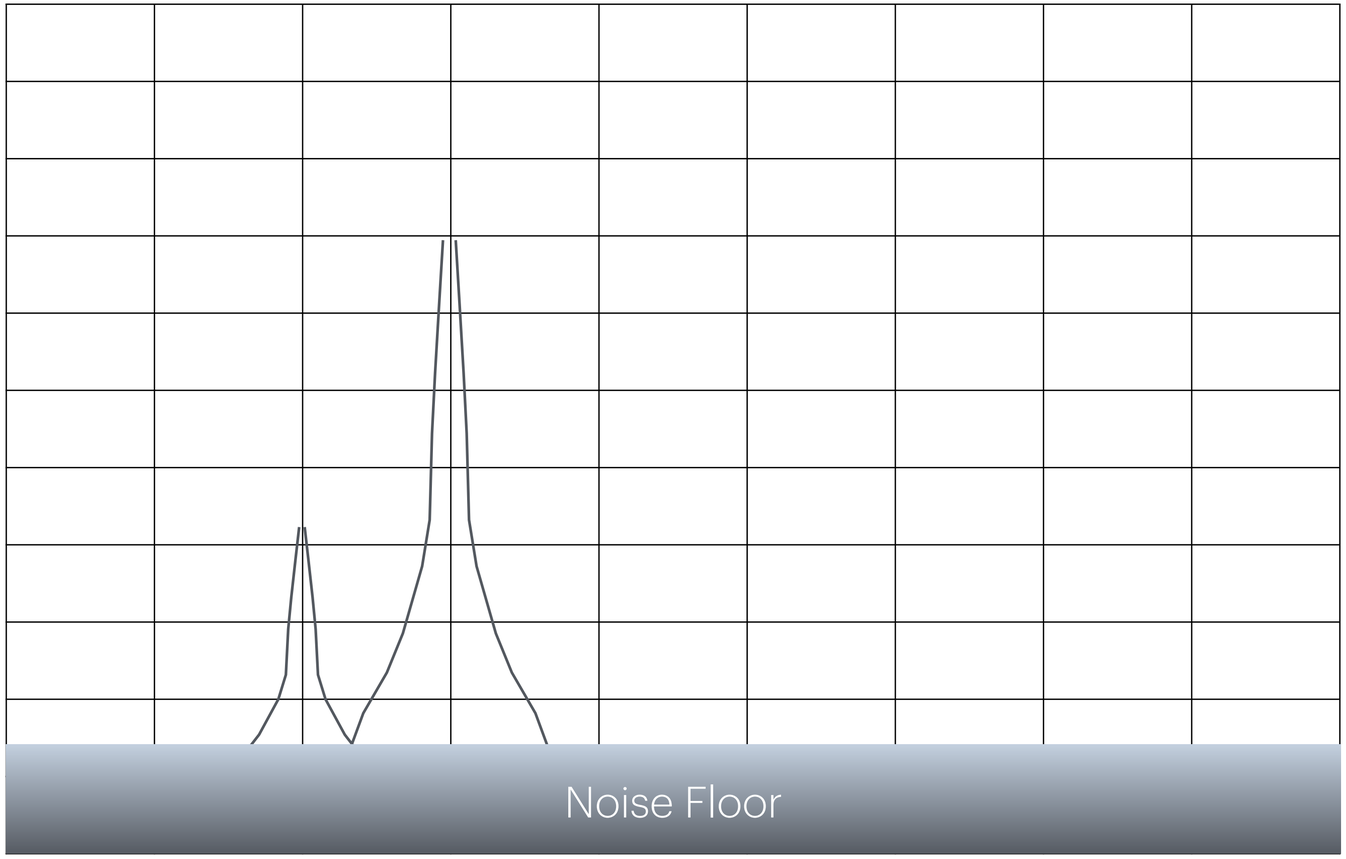
Strength of frequency components

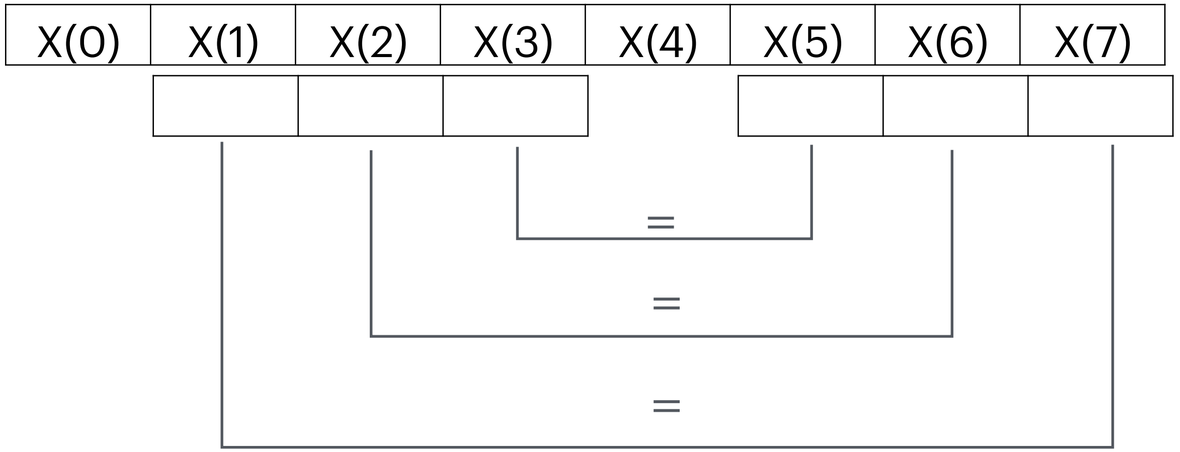


DFT

Real world signals are shaped as shown in plot.

Received signals will never have exact frequency components matching the sense capability of the sampling and DFT unit. DFT Leakage is intrinsic to signal processing.



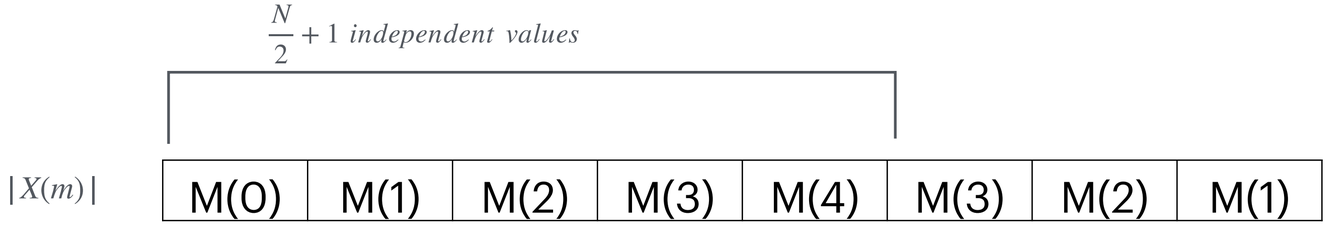


Redundancy for

$$m > \frac{N}{2}$$

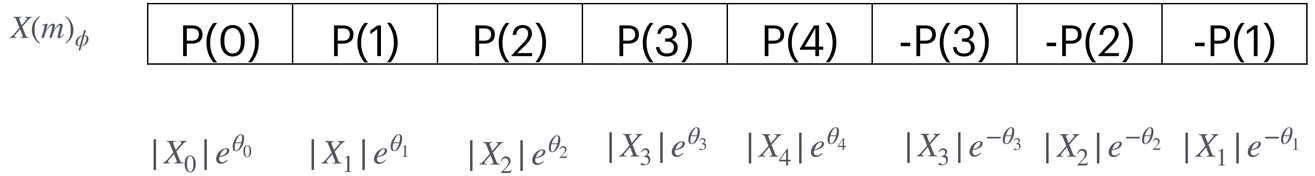
with

$$m = 1 \rightarrow m = \frac{N}{2} - 1$$



Real Part DFT - Even Symmetry

Imaginary Part DFT - Odd Symmetry



$x(n)$ $\xrightarrow{x_{real} + j\cancel{x_{img}}^{0}}$ DFT N=9 $\xrightarrow{X(m) = X(m)_{real} + jX(m)_{img}}$

X(0)	X(1)	X(2)	X(3)	X(4)	X(5)	X(6)	X(7)	X(8)

Redundancy for

$$m > \frac{N+1}{2} - 1$$

with

$$m = 0 \rightarrow m = \frac{N+1}{2} - 1$$

$\frac{N+1}{2}$ *independent values*

$|X(m)|$

M(0)	M(1)	M(2)	M(3)	M(4)	M(4)	M(3)	M(2)	M(1)

Real Part DFT - Even Symmetry

Imaginary Part DFT - Odd Symmetry

$X(m)_{\phi}$

P(0)	P(1)	P(2)	P(3)	P(4)	-P(4)	-P(3)	-P(2)	-P(1)

$|X_0|e^{\theta_0}$ $|X_1|e^{\theta_1}$ $|X_2|e^{\theta_2}$ $|X_3|e^{\theta_3}$ $|X_4|e^{\theta_4}$ $|X_3|e^{-\theta_4}$ $|X_2|e^{-\theta_3}$ $|X_1|e^{-\theta_2}$ $|X_1|e^{-\theta_1}$

$x(n)$ $\xrightarrow{x_{real_{even}} + \cancel{jx_{img}}^{\,0}}$ DFT N=8 $\xrightarrow{X(m) = X(m)_{real} + j\cancel{X(m)_{img}}^{\,0}}$

Real even produces zeroed imaginary output

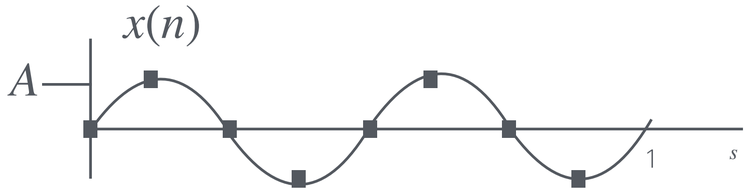
$x(n)$ $\xrightarrow{x_{real_{odd}} + \cancel{jx_{img}}^{\,0}}$ DFT N=8 $\xrightarrow{X(m) = \cancel{X(m)_{real}}^{\,0} + jX(m)_{img}}$

Real odd produces zeroed real output

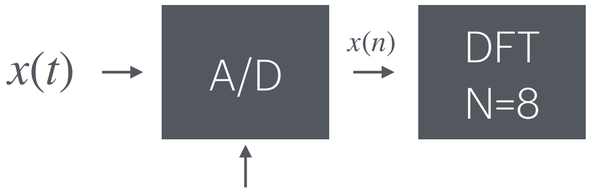
$x(n)$ $\xrightarrow{x_{real_{odd}} + jx_{img}}$ DFT N=8 $\xrightarrow{X(m) = X(m)_{real} + jX(m)_{img}}$

Complex produces Complex DFT output

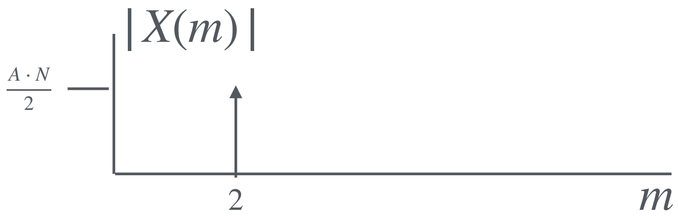
DFT Magnitude: Sinwave



2 Hz

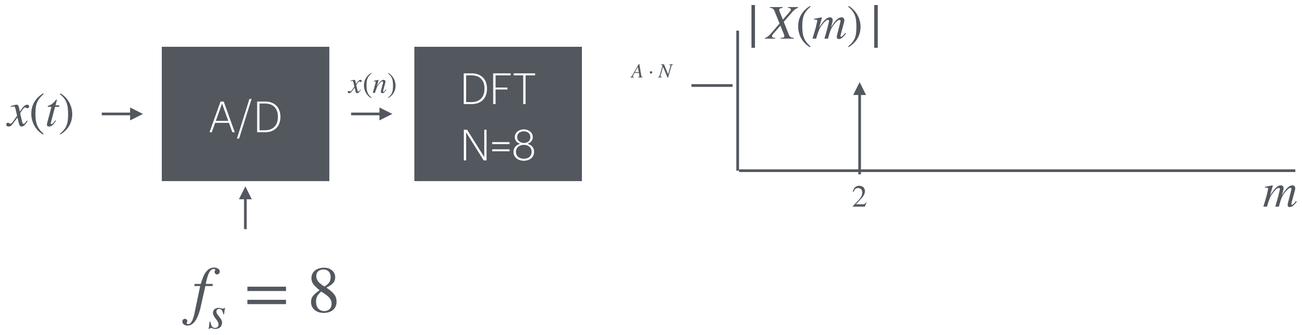


$f_s = 8$



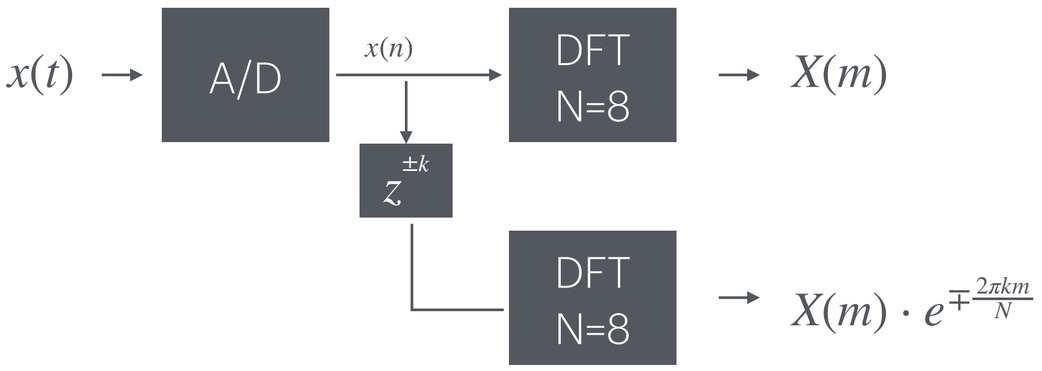
Note : Sampling frequency of 4 Hz will sample zeros and output zero energy. Make sure sampling fold frequency is more than the largest sinusoid component in input

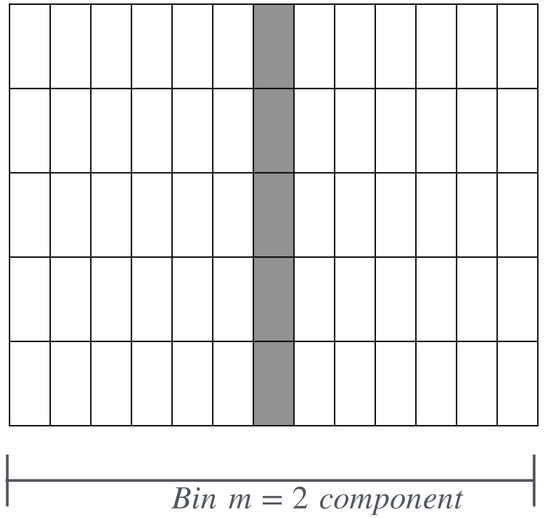
DFT Magnitude: Complex Sinusoid

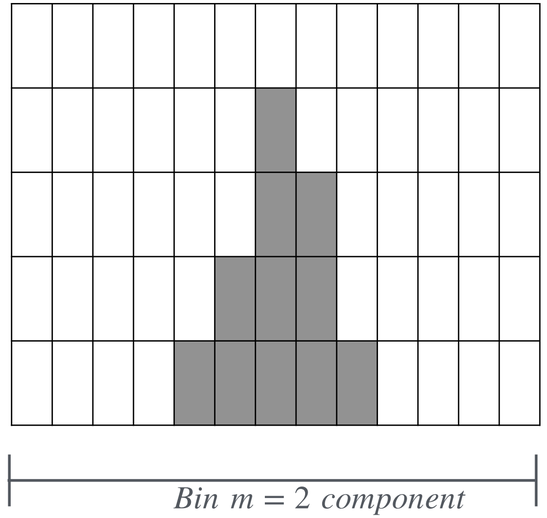


$f_s = 8$

DFT Delay-Phase Relationship





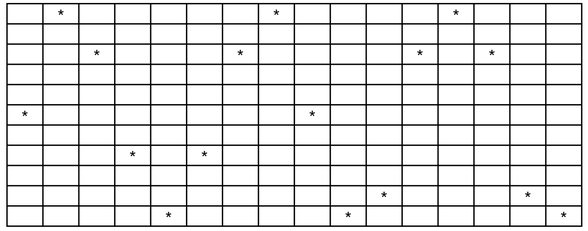


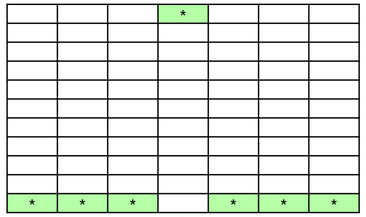
Unavoidable effect when taking DFT of real world signal. Windowing can help reduce spreading by reducing side lobe effect

Window

- Lowers side lobes while expanding main lobe width
  - Reduction in frequency resolution
- Reduces interference with other components in spectrum
  - Captures low-level signals in presence of high-level signal

$x(t) \rightarrow$ A/D $\xrightarrow{x(n)}$ DFT N=8





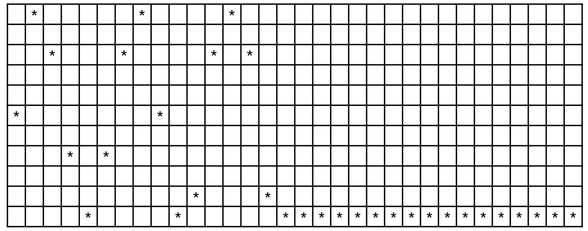
$m = 3$

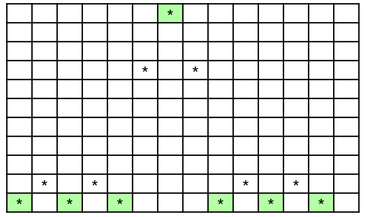
$\frac{m \cdot f_s}{16}$

$m = 3$

$x(t) \rightarrow$ A/D $\xrightarrow{x(n)}$ DFT N=16, P=8

$2x$





1(2-1) sample between init points

$m = 3$

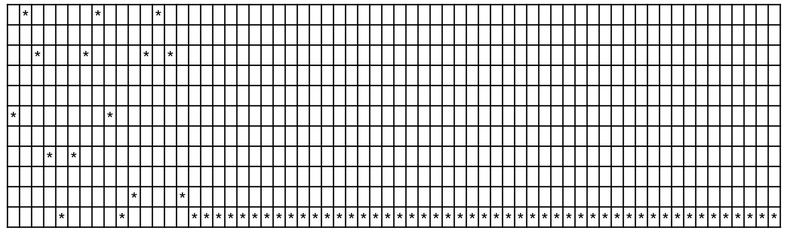
$k = 2$

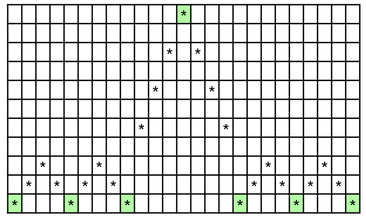
$\frac{k \cdot m \cdot f_s}{32}$

$km = 6$

$x(t) \rightarrow$ A/D $\xrightarrow{x(n)}$ DFT N=32, P=16

$4x$





3(4 -1) sample between init po

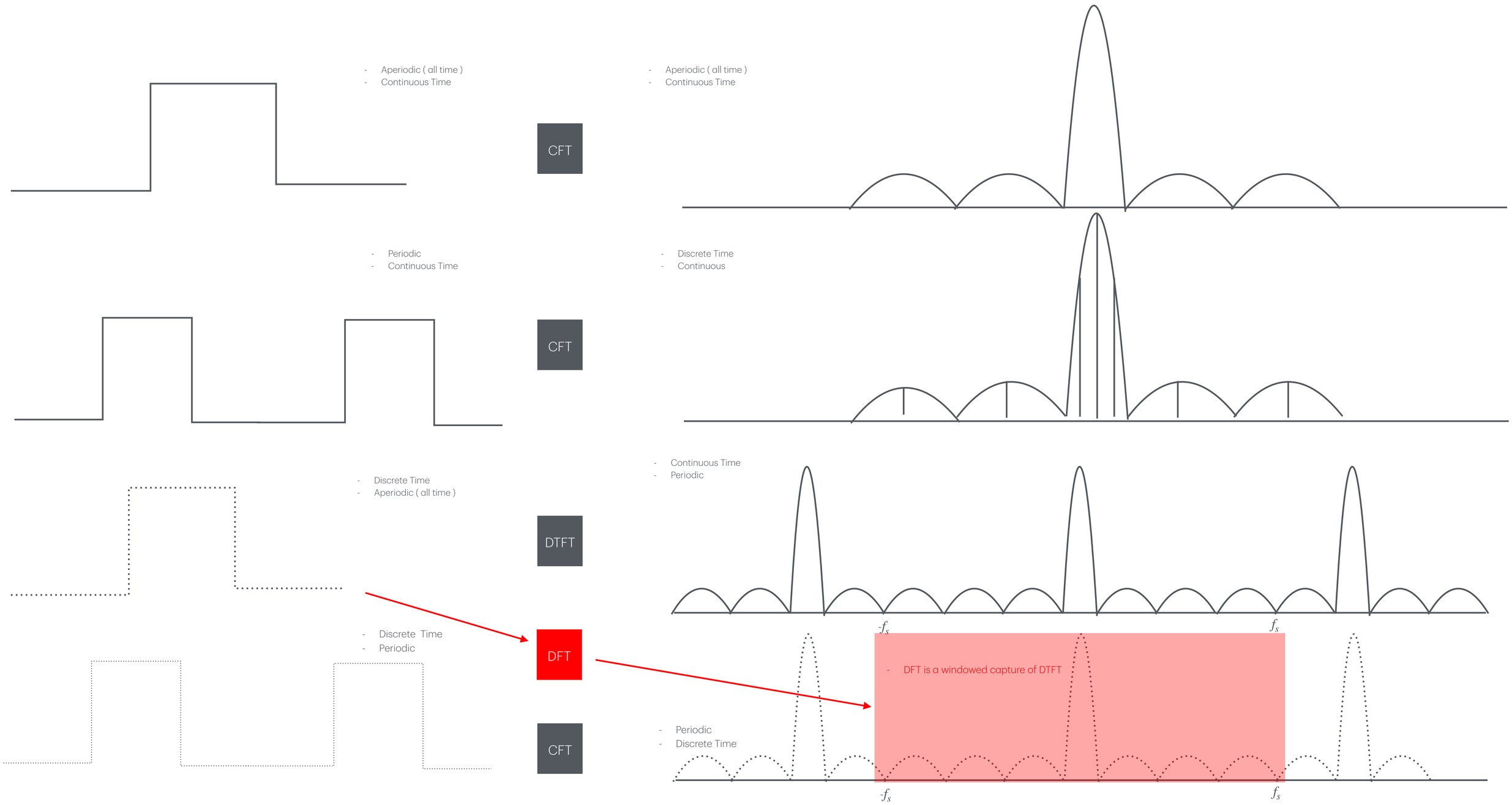
$m = 3$

$k = 4$

$\frac{k \cdot m \cdot f_s}{64}$

$km = 12$

Granularity improved


- Aperiodic (all time)
- Continuous Time
CFT
- Aperiodic (all time)
- Continuous Time
- Periodic
- Continuous Time
CFT
- Discrete Time
- Continuous
- Discrete Time
- Aperiodic (all time)
DTFT
- Continuous Time
- Periodic
$-f_s$
f_s
- Discrete Time
- Periodic
DFT
- DFT is a windowed capture of DTFT
CFT
- Periodic
- Discrete Time
$-f_s$
f_s